

Program 6

STEP 1 : Start MiniKube

```
(1RV24MC114_VENKATESHAHD@kali)-[~/mca/devops/lab/lab7]
$ minikube start --driver=docker
minikube v1.37.0 on Debian kali-rolling
Using the docker driver based on existing profile
Starting "minikube" primary control-plane node in "minikube" cluster
Pulling base image v0.0.48 ...
Restarting existing docker container for "minikube" ...
Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
Verifying Kubernetes components ...
  Using image gcr.io/k8s-minikube/storage-provisioner:v5
  Using image docker.io/kubernetesui/dashboard:v2.7.0
  Using image docker.io/kubernetesui/metrics-scraper:v1.0.8
Some dashboard features require the metrics-server addon. To enable all features please run:

    minikube addons enable metrics-server

Enabled addons: storage-provisioner, dashboard, default-storageclass
Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
```

STEP 2 : Check Minikube Status

STEP 3 : Verify Running Services

```
(1RV24MC114_VENKATESHAHD@kali)-[~/mca/devops/lab/lab7]
$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
```

STEP 4 : CREATE Deployment yaml

```
(1RV24MC114_VENKATESHAHD@kali) - [~/mca/devops/lab/lab7]
$ cat nginx-deployment.yaml
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment

spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
        - name: nginx
          image: nginx:latest
          ports:
            - containerPort: 80
```

Step 5 : Apply Deployment

STEP 6 : Check Status

```
(1RV24MC114_VENKATESHAHD@kali)-[~/mca/devops/lab/lab7]
$ kubectl apply -f nginx-deployment.yaml
deployment.apps/nginx-deployment configured
```

```
(1RV24MC114_VENKATESHAHD@kali)-[~/mca/devops/lab/lab7]
$ kubectl get deployments
```

NAME	READY	UP-TO-DATE	AVAILABLE	AGE
nginx-deployment	2/2	2	2	4d1h

```
(1RV24MC114_VENKATESHAHD@kali)-[~/mca/devops/lab/lab7]
$ kubectl get pods
```

NAME	READY	STATUS	RESTARTS	AGE
nginx-deployment-6f9664446b-kc9md	1/1	Running	0	4d1h
nginx-deployment-6f9664446b-zkwjg	1/1	Running	0	4d1h

```
(1RV24MC114_VENKATESHAHD@kali)-[~/mca/devops/lab/lab7]
$ kubectl get replicates
error: the server doesn't have a resource type "replicates"
```

```
(1RV24MC114_VENKATESHAHD@kali)-[~/mca/devops/lab/lab7]
$ kubectl get replicaset
```

NAME	DESIRED	CURRENT	READY	AGE
nginx-deployment-6f9664446b	2	2	2	4d1h

STEP 7 : Access the web app

```
(1RV24MC114_VENKATESHAHD@kali)-[~/mca/devops/lab/lab7]
$ kubectl expose deployment nginx-deployment --type=NodePort --port=80
service/nginx-deployment exposed
```

```
(1RV24MC114_VENKATESHAHD@kali)-[~/mca/devops/lab/lab7]
$ kubectl get services
```

NAME	TYPE	CLUSTER-IP	EXTERNAL-IP	PORT(S)	AGE
kubernetes	ClusterIP	10.96.0.1	<none>	443/TCP	18d
nginx-deployment	NodePort	10.109.199.196	<none>	80:30233/TCP	17s
nginx-service	NodePort	10.99.253.172	<none>	80:30813/TCP	4d1h

```
(1RV24MC114_VENKATESHAHD@kali)-[~/mca/devops/lab/lab7]
$ minikube service nginx-deployment
```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-deployment	80	http://192.168.49.2:30233

Opening service default/nginx-deployment in default browser...

```
(1RV24MC114_VENKATESHAHD@kali)-[~/mca/devops/lab/lab7]
$ Opening in existing browser session.
```

STEP 8 : DELETE all pods

kubectl delete pods —all